Complementary Material to Bayesian Adaptive Portfolio Optimisation Exploiting Fully Probabilistic Design

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Abstract

This document complements Sec. 4 of the paper. It documents the experimental protocol in full (data, universe, dates, cost model, metrics, benchmarks, seeds), gives the end-to-end algorithm and the complete list of user-set quantities, and reports the results that the paper only summarises: the paired tests against every benchmark, the component ablation, the sensitivity to the user-set quantities, the behaviour of the elicited risk weight, the synthetic stress test, the three additional market windows and the live test. It also lists the implementation errors of the previous version of the manuscript and their consequences. Section and equation numbers without the prefix S refer to the paper; everything numbered Sn is in this document. All tables and figures are regenerated from the released code (`https://github.com/ProchTomas/ResearchProject`) by a single script.

S1. Data and Universe

The universe is built from the S&P 500 constituents: of the 100 members with the highest average daily return in 2021, we keep the 20 with the lowest pair-wise correlation (Fig. S5), namely MRNA, COP, MOH, AXON, WST, EQT, AZO, DXCM, DELL, EXR, ENPH, BG, CRWD, EPAM, OXY, TGT, TSLA, CF, ANET and TSCO. Returns are simple daily returns of dividend- and split-adjusted closing prices. The screen on 2021 returns precedes the test period, but it is a screen on past performance and the resulting universe is therefore not neutral; Sec. S9 reports the same experiments on periods before and after the screen, and every table contains the S&P 500, so that the effect of the universe stays visible.

Each experiment uses 750 trading days per portfolio: 125 days for the optimised prior (Sec. 2.1 of the paper), 125 days for the structure estimation (Sec. 2.2), and 500 test days, on which all reported metrics are computed. For the main window these are 4 November 2022 to 31 October 2025, with the test period starting on 3 November 2023. We draw 200 portfolios of 8 to 12 distinct tickers uniformly at random with `numpy` generator seed 0 (100 portfolios for the ablations, the sensitivity study and the additional windows, seed 0 as well) and run every strategy, benchmark and ablation on exactly the same portfolios and the same days, so that all comparisons are paired and none of them is affected by the sampling.

The candidate regressor vector $\bar{z}_t$ contains 10 lagged returns of every asset of the portfolio, three differences of exponential moving averages with spans $(5, 10)$, $(5, 20)$ and $(1, 30)$, three sines of periods 10, 30 and 90 trading days, and an intercept; for a ten-asset portfolio this makes 137 candidates. It uses returns up to $t - 1$ only.

S2. Costs, Metrics and Testing

Every strategy starts from the uniform allocation and pays a proportional cost on the value actually traded,

$$c_t = 0.002 \sum_{i=1}^{N} |a_{i,t} - \tilde{a}_{i,t-1}|, \qquad \tilde{a}_{i,t-1} = \frac{a_{i,t-1}(1 + y_{i,t-1})}{1 + a_{t-1}^{\top} y_{t-1}}, \tag{S1}$$

so that the drift of the weights caused by the realised returns is not charged, and wealth evolves as $W_{t+1} = W_t(1 + a_t^{\top} y_t)(1 - c_t)$. The same routine prices every strategy of this study, the buy-and-hold ones included. In the previous version of the paper the uniform portfolio was charged no costs, which is the main reason why its ranking changes here.

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The reported metrics are the total return W_T/W_0, the annualised return and volatility of the daily net returns (252 days), the annualised Sharpe ratio with a zero risk-free rate, the maximum drawdown, the Calmar ratio, the mean daily turnover $\sum_i |a_{i,t} - \tilde{a}_{i,t-1}|$, and the total cost paid as a fraction of the initial wealth. Because all strategies share the portfolios, differences are formed per portfolio and tested with the Wilcoxon signed-rank test, with a bootstrap 95% confidence interval (10^4 resamples) of the mean difference, and with the win rate, the fraction of portfolios in which our policy is the better one. The p-values carry no multiplicity correction and are to be read as descriptive.

S3. Benchmarks

On identical portfolios, days and cost model we compare: equal weights rebalanced daily; equal weights bought and held, which is the "Uniform Start" policy of the previous version; inverse volatility on a 60-day window; minimum-variance and long-only mean-variance with risk aversion 5, both with a Ledoit–Wolf shrunk covariance on 120 and 250 days, the mean-variance one also in a cost-aware version that penalises the trade with our **D**; an ARX-conditional mean-variance, which feeds the regressors selected by our own structure estimation into a rolling ridge regression and then into the same optimiser, and so separates the information contained in z_t from our Bayesian learning and from the FPD loss; and buy-and-hold of the S&P 500 (SPY, total return), a different and much larger universe, included because practitioners use it as the reference.

Table S1 repeats the means of the main window with all metrics, Table S2 gives the paired comparison of the Risk-Averse policy (RA) against each benchmark on the main window, Table S3 the medians over the portfolios, Fig. S1 the median wealth paths, and Table S4 the same paired comparison for the legacy softmax policy, which is the one the tables of the previous version of the manuscript actually described.

S4. End-to-End Algorithm and User-Set Quantities

Algorithm 1 (Fig. S2) summarises the complete procedure as implemented in the released code, and Table S5 lists every user-set quantity, its role, its value and how it was chosen. One portfolio costs, on a single core and without compiled code, 2.6 s for the prior and for the 500-iteration genetic search over 2^{137} structures, which keeps 57 of the 137 candidates on average, and 0.1 s per trading day of the on-line loop. The

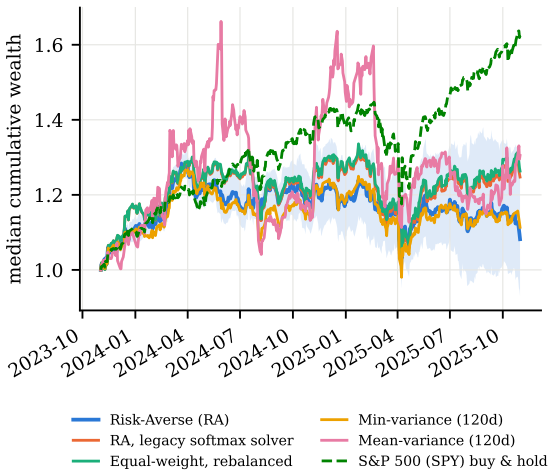


Figure S1: Median cumulative wealth over the 200 portfolios; the band is the interquartile range of RA. The S&P 500 is the same for all portfolios.

whole study, about 3000 portfolio-years, ran in under two hours on 20 cores; Table S6 reports the measured times of all variants.

S5. Component Ablation

One component is removed or replaced at a time, everything else identical: no forgetting ($\alpha = 0, \beta = 1$); exponential forgetting with factor 0.99; the risk-neutral loss, that is the FPD risk term dropped; no structure estimation, all 137 candidates kept; a flat prior ($\mathbf{V}_0 = 10^{-6}\mathbf{I}$, $\Delta_0 = \rho + 2$) in place of the optimised one; the receding horizon $h = 5$ without loss-to-go propagation; Thompson sampling of θ in place of the point estimate; and the constrained-QP solver of Sec. 3.2 against the legacy softmax heuristic. Table S7 reports the means and Table S8 the paired tests, both on 100 portfolios.

Essential. Removing the FPD risk term is catastrophic: the policy follows the point forecast, turns the portfolio over 1.5 times per day and loses 72% of the capital. Thompson sampling of θ (−56%) and dropping the structure estimation (−49%) do the same, and for the same reason: they widen or distort the predictive density that the loss then takes at face value.

Inert. The corrected SMALBE solver and the SLSQP reference give identical allocations, the largest weight difference over all steps being $3 \cdot 10^{-7}$, and $h = 5$ reproduces $h = 1$ to $7 \cdot 10^{-6}$, because at $\omega = 10^3$ the one-step loss dominates the loss-to-go. Replacing the data-based forgetting by a plain Bayesian update changes nothing

strategy	total return	ann. return	ann. vol.	Sharpe (ann.)	max DD	Calmar	turnover/day	cost paid
Risk-Averse (RA)	1.114	5.0%	19.2%	0.34	-25.5%	0.33	14.9%	14.9%
RA, legacy softmax solver	1.257	11.9%	21.0%	0.63	-27.4%	0.47	2.9%	2.9%
Equal-weight, rebalanced	1.273	12.6%	20.9%	0.66	-27.1%	0.50	1.6%	1.6%
Equal-weight, buy & hold	1.357	16.3%	22.1%	0.77	-28.8%	0.60	0.0%	0.0%
Inverse volatility (60d)	1.198	9.3%	19.0%	0.55	-26.4%	0.39	2.3%	2.3%
Min-variance (120d)	1.128	6.0%	17.7%	0.41	-24.6%	0.30	2.8%	2.8%
Min-variance (250d)	1.181	8.6%	17.3%	0.56	-23.0%	0.43	2.0%	2.0%
Mean-variance (120d)	1.277	12.4%	37.2%	0.47	-41.6%	0.33	15.3%	15.3%
Mean-variance (250d)	1.567	24.0%	37.0%	0.69	-39.8%	0.67	8.6%	8.6%
Mean-variance, cost-aware (120d)	1.362	16.1%	36.2%	0.56	-41.8%	0.42	1.1%	1.1%
ARX-conditional mean-variance	0.732	-15.2%	18.7%	-0.83	-40.6%	-0.33	43.6%	43.6%
S&P 500 (SPY) buy & hold	1.624	27.7%	16.3%	1.58	-18.8%	1.48	–	–

Table S1: Benchmark comparison on the 500 test days 2023-11-03 to 2025-10-31, all strategies on the same random 8–12 stock portfolios, 0.2% proportional transaction cost (mean over 200 portfolios).

benchmark	ΔSharpe	95% CI	p	win rate	Δmax DD	p	Δtotal ret.	p
RA, legacy softmax solver	-0.29	[-0.35, -0.22]	<0.001	28%	+2.0%	<0.001	-0.143	<0.001
Equal-weight, rebalanced	-0.32	[-0.38, -0.26]	<0.001	26%	+1.7%	0.002	-0.159	<0.001
Equal-weight, buy & hold	-0.43	[-0.49, -0.37]	<0.001	20%	+3.3%	<0.001	-0.243	<0.001
Inverse volatility (60d)	-0.21	[-0.27, -0.15]	<0.001	30%	+0.9%	0.029	-0.084	<0.001
Min-variance (120d)	-0.07	[-0.13, -0.02]	0.008	44%	-0.9%	0.152	-0.014	0.089
Min-variance (250d)	-0.22	[-0.27, -0.16]	<0.001	34%	-2.5%	<0.001	-0.067	<0.001
Mean-variance (120d)	-0.13	[-0.20, -0.06]	<0.001	38%	+16.1%	<0.001	-0.164	<0.001
Mean-variance (250d)	-0.35	[-0.42, -0.27]	<0.001	24%	+14.4%	<0.001	-0.454	<0.001
Mean-variance, cost-aware (120d)	-0.22	[-0.28, -0.15]	<0.001	34%	+16.4%	<0.001	-0.248	<0.001
ARX-conditional mean-variance	+1.17	[+1.09, +1.25]	<0.001	100%	+15.1%	<0.001	+0.382	<0.001
S&P 500 (SPY) buy & hold	-1.24	[-1.32, -1.17]	<0.001	0%	-6.7%	<0.001	-0.511	<0.001

Table S2: Paired comparison of the Risk-Averse policy against every benchmark across portfolios: mean difference of the annualised Sharpe ratio with bootstrap 95% CI, Wilcoxon signed-rank p-value, fraction of portfolios in which RA is better; the same for maximum drawdown (positive = smaller drawdown) and total return.

strategy	total return	ann. return	ann. vol.	Sharpe (ann.)	max DD	Calmar	turnover/day	cost paid
Risk-Averse (RA)	1.082	4.0%	19.1%	0.30	-25.1%	0.18	14.6%	14.6%
RA, legacy softmax solver	1.247	11.8%	20.8%	0.65	-26.9%	0.44	2.8%	2.8%
Equal-weight, rebalanced	1.263	12.5%	20.8%	0.66	-26.6%	0.48	1.6%	1.6%
Equal-weight, buy & hold	1.362	16.9%	22.0%	0.84	-28.0%	0.63	0.0%	0.0%
Inverse volatility (60d)	1.196	9.4%	18.8%	0.57	-26.2%	0.36	2.3%	2.3%
Min-variance (120d)	1.114	5.6%	17.5%	0.40	-24.3%	0.25	2.8%	2.8%
Min-variance (250d)	1.165	8.0%	17.2%	0.54	-23.0%	0.34	2.0%	2.0%
Mean-variance (120d)	1.305	14.4%	37.6%	0.54	-41.1%	0.36	15.4%	15.4%
Mean-variance (250d)	1.477	21.7%	38.1%	0.72	-39.3%	0.60	8.4%	8.4%
Mean-variance, cost-aware (120d)	1.336	15.7%	36.8%	0.58	-42.3%	0.39	1.1%	1.1%
ARX-conditional mean-variance	0.735	-14.4%	18.4%	-0.75	-39.9%	-0.37	39.1%	39.1%
S&P 500 (SPY) buy & hold	1.624	27.7%	16.3%	1.58	-18.8%	1.48	–	–

Table S3: Benchmark comparison, medians (median over 200 portfolios).

measurable (Sharpe +0.04, $p = 0.13$), and the forgetting prior (α_0, β_0) moves the realised weights but no metric. Fig. S3 explains this: the optimiser holds γ, the weight of the prior, between 0.5 and 0.8 on almost every day, so the statistic is pulled back towards $\mathbf{V}_0$ faster than the data accumulate in it.

A trap. The flat prior looks like the best variant of all (Sharpe 0.72) for a trivial reason: with $\mathbf{V}_0 = 10^{-6}\mathbf{I}$ the forecasts vanish, the predictive covariance is isotropic, and the constrained minimiser is exactly $1/N$; the largest deviation from the uniform allocation over all portfolios and days is $1.9 \cdot 10^{-11}$. The variant is the equal-weight portfolio under another name, and we report it as such.

S6. Sensitivity to the User-Set Quantities

Table S9 varies the tuned quantities on 100 portfolios and Table S10 tests the differences. Of the three, only the prior shrinkage μ matters, and the value 0.9 used in the paper is not its best one: $\mu = 0.7$ keeps 15 of the

benchmark	ΔSharpe	95% CI	p	win rate	Δmax DD	p	Δtotal ret.	p
Risk-Averse (RA)	+0.29	[+0.22, +0.35]	<0.001	72%	-2.0%	<0.001	+0.143	<0.001
Equal-weight, rebalanced	-0.03	[-0.03, -0.03]	<0.001	0%	-0.3%	<0.001	-0.016	<0.001
Equal-weight, buy & hold	-0.14	[-0.15, -0.13]	<0.001	4%	+1.3%	<0.001	-0.100	<0.001
Inverse volatility (60d)	+0.08	[+0.06, +0.09]	<0.001	75%	-1.0%	<0.001	+0.059	<0.001
Min-variance (120d)	+0.21	[+0.19, +0.24]	<0.001	85%	-2.9%	<0.001	+0.129	<0.001
Min-variance (250d)	+0.07	[+0.04, +0.10]	<0.001	59%	-4.5%	<0.001	+0.076	<0.001
Mean-variance (120d)	+0.16	[+0.12, +0.21]	<0.001	68%	+14.2%	<0.001	-0.021	0.349
Mean-variance (250d)	-0.06	[-0.12, +0.00]	0.028	44%	+12.4%	<0.001	-0.311	<0.001
Mean-variance, cost-aware (120d)	+0.07	[+0.03, +0.12]	0.002	60%	+14.4%	<0.001	-0.105	<0.001
ARX-conditional mean-variance	+1.46	[+1.37, +1.55]	<0.001	100%	+13.2%	<0.001	+0.524	<0.001
S&P 500 (SPY) buy & hold	-0.96	[-1.00, -0.91]	<0.001	0%	-8.7%	<0.001	-0.368	<0.001

Table S4: Paired comparison of the legacy (softmax) policy against every benchmark.

Algorithm 1 Bayesian adaptive portfolio optimisation

Input: returns $y_t \in \mathbb{R}^N$, candidate regressors $\bar{z}_t \in \mathbb{R}^{\bar{p}}$, cost matrix $\mathbf{D}$, prior weights (α_0, β_0), shrinkage μ, warm-up lengths $\bar{t}, T$.

Off-line (days $1 \dots T$*)*

1. Collect $\tilde{\mathbf{V}}_{\bar{t}} = \sum_{\tau \le \bar{t}} (y_\tau^\top, \bar{z}_\tau^\top)^\top (y_\tau^\top, \bar{z}_\tau^\top)$; solve the scalar equation of Sec. 2.1 for Δ_0; form $(\mathbf{V}_0, \mathbf{G}_0)$.
2. Accumulate $\mathbf{V}_T = \mathbf{V}_0 + \sum_{\bar{t} < \tau \le T} (\cdot)(\cdot)^\top$; run the genetic algorithm of Sec. 2.2 on $\ln l(u)$ (15); keep the regressors $z_t \subset \bar{z}_t$ of the maximiser u^*; restrict $\mathbf{V}_0$ to (y_t, z_t) and recompute its square root $\mathbf{G}_0$.
3. Set $\mathbf{G}_T = \mathbf{G}_0$, $\Delta_T = \Delta_0$, $a_T = \mathbf{1}/N$, $\mathbf{H}_T = \epsilon \mathbf{I}$.

On-line (each trading day $t > T$*)*

4. **Predict:** $\hat{\mathbf{P}} = \mathbf{G}_{zy}^\top \mathbf{G}_z^{-\top}$, $\hat{y}_t = \hat{\mathbf{P}}^\top z_t$, $\hat{\mathbf{\Sigma}} = \mathbf{G}_y \mathbf{G}_y^\top / \Delta_{t-1}$ (14).
5. **Elicit risk weight:** $\omega^* = \arg\min_{\omega > 0}$ of (20) given $\mathbf{D}$, $\hat{\mathbf{\Sigma}}$, a_{t-1}; build $\mathbf{Q}_{t-1}$ (21).
6. **Design:** one Riccati step of Sec. 3.2 from $\mathbf{H}_{t-1}$ gives $(\mathbf{H}_a, \mathbf{H}_s)$; solve the constrained QP $\min_{a \in \Omega_{BE}} \|\mathbf{H}_a a + \mathbf{H}_s s_{t-1}\|^2$ (SMALBE, or any simplex-QP solver); output a_t.
7. **Trade** to a_t, pay $\mathbf{D}$-proportional costs, observe y_t.
8. **Learn:** minimise the forgetting criterion $F(\alpha, \beta, \gamma)$ of Sec. 2.3 on the simplex (Nelder–Mead in the softmax parametrisation); update $\mathbf{G}_t$ by the orthogonal rotation of Sec. 2.3 and $\Delta_t = (\alpha^* + \beta^*)\Delta_{t-1} + \beta^* + \gamma^* \Delta_0$.
9. **Propagate loss-to-go:** $\mathbf{H}_t = \phi^* \mathbf{H}_{t-1} + (1 - \phi^*) \mathbf{H}_t^{\text{new}}$ with ϕ^* minimising the KLD criterion of Sec. 3.2.

Figure S2: The complete procedure (Algorithm 1); step numbers refer to the released implementation.

quantity	role	value
μ	prior shrinkage, Sec. 2.1	0.9
α_0, β_0	prior forgetting weights, Sec. 2.3	0.09, 0.5
ϕ_0	prior weight of loss-to-go mixing	0.1
$\bar{t}, T$	prior / structure warm-up [days]	125, 125
h	receding horizon	1
$\mathbf{D}$	transaction cost per unit traded	$0.002\,\mathbf{I}$
$\bar{z}_t$	candidate regressors	10 lags, 3 EMA differences, 3 sines, intercept
GA	batch, p_{mut}, α_{decay}, iterations	8, 0.5, 0.992, 500
$\omega_{\max}$	upper bound of the risk-weight search	10^3

Table S5: All user-set quantities of the experiments. μ, α_0, β_0 were chosen by a 2^k design on 2021–2022 data, disjoint from the test period; the remaining values are conventional defaults that were not tuned.

137 candidates instead of 57 and gains 0.18 in Sharpe ratio ($p < 0.001$), still below the equal-weight portfolio, while $\mu = 0.95$ keeps 96 of them and loses 0.11. The forgetting prior (α_0, β_0) changes the realised weights but no metric beyond the noise level.

S7. The Risk Weight ω

The elicitation of Sec. 3.1 returns the upper end $\omega_{\max} = 10^3$ of the searched bracket in all of the 10^5 decision steps of the study. This is not a numerical accident. The inner minimum in Eq. (20) of the paper is minus the logarithm of a Gaussian integral that decreases monotonically in ω, so the outer maximisation has no finite solution, and the necessary condition for an extreme is a sum of terms that are positive for $\mathbf{D} \succ \mathbf{O}$, $\hat{\mathbf{\Sigma}} \succeq \mathbf{O}$ and $\omega > 0$, hence has no root; the implementation minimises the left-hand side of that condition on

variant	regressors selected	setup [s]	time / step [s]	fallbacks
Risk-Averse (RA)	57 / 137	2.6	0.10	0
RA, legacy softmax solver	57 / 137	2.6	0.09	0
RA, SMALBE solver	57 / 136	2.5	0.24	0
Risk-Neutral loss	57 / 136	2.6	0.09	0
RA, no forgetting	57 / 136	2.6	0.08	0
RA, exp. forgetting 0.99	57 / 136	2.6	0.08	0
RA, no structure est.	136 / 136	0.0	0.37	0
RA, flat prior	131 / 136	6.7	0.23	0
RA, horizon 5	57 / 136	2.8	0.04	0
RA, Thompson sampling	57 / 136	2.8	0.10	0

Table S6: Computational cost per portfolio (means): regressors kept by the structure estimation out of the full set, prior + structure estimation time, and wall time per trading day of the online loop (single core).

strategy	total return	ann. return	ann. vol.	Sharpe (ann.)	max DD	Calmar	turnover/day	cost paid
Risk-Averse (RA)	1.126	5.6%	19.4%	0.37	-25.5%	0.36	14.8%	14.8%
RA, legacy softmax solver	1.287	13.3%	21.0%	0.68	-26.6%	0.53	2.9%	2.9%
RA, SMALBE solver	1.126	5.6%	19.4%	0.37	-25.5%	0.36	14.8%	14.8%
Risk-Neutral loss	0.279	-50.5%	47.5%	-1.35	-80.8%	-0.60	155.4%	155.4%
RA, no forgetting	1.128	6.0%	17.9%	0.41	-22.6%	0.37	14.8%	14.8%
RA, exp. forgetting 0.99	1.032	1.3%	17.7%	0.15	-26.1%	0.12	20.1%	20.1%
RA, no structure est.	0.512	-31.6%	40.7%	-0.82	-64.2%	-0.44	106.8%	106.8%
RA, flat prior	1.304	14.0%	21.0%	0.72	-26.3%	0.57	1.6%	1.6%
RA, horizon 5	1.126	5.6%	19.4%	0.37	-25.5%	0.36	14.8%	14.8%
RA, Thompson sampling	0.442	-37.1%	27.5%	-1.60	-63.4%	-0.50	118.7%	118.7%

Table S7: Ablation study: each row removes or replaces one component of the Risk-Averse policy (mean over 100 portfolios).

benchmark	ΔSharpe	95% CI	p	win rate	Δmax DD	p	Δtotal ret.	p
RA, legacy softmax solver	-0.32	[-0.40, -0.23]	<0.001	26%	+1.1%	0.138	-0.161	<0.001
RA, SMALBE solver	-0.00	[-0.00, +0.00]	0.853	51%	+0.0%	0.931	-0.000	0.915
Risk-Neutral loss	+1.72	[+1.58, +1.87]	<0.001	99%	+55.4%	<0.001	+0.847	<0.001
RA, no forgetting	-0.04	[-0.08, +0.00]	0.130	47%	-2.9%	<0.001	-0.002	0.667
RA, exp. forgetting 0.99	+0.21	[+0.15, +0.28]	<0.001	74%	+0.7%	0.077	+0.095	<0.001
RA, no structure est.	+1.19	[+1.04, +1.35]	<0.001	94%	+38.8%	<0.001	+0.614	<0.001
RA, flat prior	-0.35	[-0.43, -0.26]	<0.001	23%	+0.9%	0.238	-0.178	<0.001
RA, horizon 5	-0.00	[-0.00, +0.00]	0.100	42%	-0.0%	0.022	-0.000	0.267
RA, Thompson sampling	+1.97	[+1.80, +2.15]	<0.001	100%	+38.0%	<0.001	+0.684	<0.001

Table S8: Paired comparison of the Risk-Averse policy against its ablations.

strategy	total return	ann. return	ann. vol.	Sharpe (ann.)	max DD	Calmar	turnover/day	cost paid
Risk-Averse (RA)	1.126	5.6%	19.4%	0.37	-25.5%	0.36	14.8%	14.8%
RA, $\mu = 0.5$	1.150	7.0%	18.4%	0.45	-22.0%	0.42	16.9%	16.9%
RA, $\mu = 0.7$	1.190	8.8%	18.4%	0.55	-21.5%	0.53	10.9%	10.9%
RA, $\mu = 0.95$	1.096	4.2%	19.7%	0.30	-26.5%	0.25	18.4%	18.4%
RA, $(\alpha_0, \beta_0) = (0.2, 0.2)$	1.127	5.6%	19.4%	0.37	-25.5%	0.36	14.8%	14.8%
RA, $(\alpha_0, \beta_0) = (0.3, 0.6)$	1.122	5.5%	19.0%	0.37	-24.8%	0.37	14.9%	14.9%
RA, $(\alpha_0, \beta_0) = (0.5, 0.45)$	1.119	5.4%	18.8%	0.37	-24.5%	0.35	15.2%	15.2%
RA, $\omega_{\max} = 1$	1.305	14.0%	21.0%	0.72	-26.3%	0.57	1.6%	1.6%
RA, $\omega_{\max} = 10$	0.266	-50.5%	38.7%	-1.72	-79.1%	-0.63	159.7%	159.7%
RA, $\omega_{\max} = 100$	0.541	-27.5%	23.3%	-1.31	-55.4%	-0.48	92.4%	92.4%
RA, ω: unit KLD	0.353	-41.8%	29.5%	-1.76	-70.3%	-0.58	132.2%	132.2%
RA, ω: match prediction	0.367	-40.4%	27.4%	-1.82	-68.6%	-0.58	127.8%	127.8%
RA, $\omega = 40$	0.358	-41.3%	28.4%	-1.80	-69.7%	-0.58	131.5%	131.5%

Table S9: Sensitivity to the user-set hyperparameters (prior shrinkage μ, prior forgetting weights) (mean over 100 portfolios).

the bracket and necessarily returns an end point of it. Capping the bracket isolates the mechanism. At

benchmark	ΔSharpe	95% CI	p	win rate	Δmax DD	p	Δtotal ret.	p
RA, $\mu = 0.5$	-0.08	[-0.13, -0.03]	0.005	40%	-3.5%	<0.001	-0.024	0.030
RA, $\mu = 0.7$	-0.18	[-0.23, -0.13]	<0.001	21%	-3.9%	<0.001	-0.063	<0.001
RA, $\mu = 0.95$	+0.07	[+0.02, +0.12]	0.005	65%	+1.1%	0.006	+0.030	0.010
RA, $(\alpha_0, \beta_0) = (0.2, 0.2)$	-0.00	[-0.00, -0.00]	<0.001	36%	+0.0%	0.201	-0.001	<0.001
RA, $(\alpha_0, \beta_0) = (0.3, 0.6)$	-0.00	[-0.02, +0.01]	0.003	68%	-0.7%	0.341	+0.004	0.003
RA, $(\alpha_0, \beta_0) = (0.5, 0.45)$	+0.00	[-0.02, +0.02]	0.570	53%	-0.9%	<0.001	+0.007	0.196
RA, $\omega_{\max} = 1$	-0.35	[-0.44, -0.26]	<0.001	23%	+0.9%	0.236	-0.179	<0.001
RA, $\omega_{\max} = 10$	+2.09	[+1.95, +2.23]	<0.001	99%	+53.6%	<0.001	+0.860	<0.001
RA, $\omega_{\max} = 100$	+1.68	[+1.60, +1.76]	<0.001	100%	+29.9%	<0.001	+0.585	<0.001
RA, ω: unit KLD	+2.13	[+2.02, +2.24]	<0.001	99%	+44.9%	<0.001	+0.773	<0.001
RA, ω: match prediction	+2.19	[+2.08, +2.30]	<0.001	100%	+43.1%	<0.001	+0.759	<0.001
RA, $\omega = 40$	+2.17	[+2.07, +2.28]	<0.001	100%	+44.2%	<0.001	+0.768	<0.001

Table S10: Paired comparison of the default hyperparameters against the alternatives.

$\omega_{\max} = 1$ the search returns $\omega \approx 10^{-5}$, the return term leaves the loss, $a_t = a_{t-1}$ becomes optimal, and the policy reproduces the rebalanced equal-weight portfolio to three decimals (total return 1.305, Sharpe 0.72). At $\omega_{\max} = 10$ and 100 the forecasts dominate the variance term, the policy takes corner positions in single stocks, and the capital falls by 73% and 46% (Sharpe -1.7 and -1.3). Three principled elicitations used in place of the bound, namely a unit Kullback–Leibler divergence between the modelled and the ideal behaviour, a mean shift that matches the predicted return, and a fixed $\omega = 40$, all lie between 27 and 91 in 80% of the steps and lose 63 to 65% (Sharpe -1.8). Only $\omega = 10^3$, where the forecasts are effectively scaled out of the decision, leaves a usable policy, and it is then the variance term alone that allocates.

The forecasts explain why. Over the test period the one-step predictions have the same standard deviation as the realised returns (0.0267 against 0.0269), a correlation of -0.01 with them, a sign accuracy of 49.7%, and twice the squared error of predicting zero. A loss that takes such a forecast at face value trades on noise, so no analytical form of ω^* can repair the design: the point forecast has to be shrunk, or the posterior parameter uncertainty integrated into the design, before it enters the loss. The paper therefore withdraws the derivation of ω^* from its claims and treats ω as a user-set bound.

S8. Stress Tests

Fig. S4 uses ARX data with known parameters (4 assets, 6 regressors, 700 days), two one-day crashes of -12% on all assets, and a sign flip of all regression coefficients at day 550. On the crash days the data-based forgetting sets $\beta = 0$ and $\gamma \approx 1$: the record is discarded, the statistic reverts to the prior, and the estimation error does not move, whereas both the plain Bayesian update and exponential forgetting absorb the outlier. This is the robustness claimed in Sec. 2.3 of the paper, and it holds.

The same figure shows its price. During the stationary phase the estimation error stays at its prior level (0.72) while the two other learners improve to 0.5, and after the parameter change the weights return at once to their prior values and the model does not re-learn, while exponential forgetting halves the error within 150 days. The mechanism protects against outliers by not learning, which is also why removing it costs nothing in Table S7. On real data the lower panel of Fig. S3 shows the median drawdown around the tariff shock of 2–9 April 2025: the policy with and without forgetting behave alike, for the same reason.

S9. Other Market Windows

The main window is a bull market. Table S11 repeats the comparison on three further 500-day windows with the same universe, protocol and hyperparameters and 100 portfolios each: 2019–2020 with the COVID crash, 2021–2022 with the bear market, and September 2024 to September 2026, which lies after all data used to design and tune the method. The first two use the 20 stocks selected by the 2021 screen on periods preceding it and therefore carry its bias; the third does not.

Three facts hold in every window. RA stays below the rebalanced equal-weight portfolio in Sharpe ratio (by 0.2 to 0.8, $p < 0.001$), paying 11 to 15% of its wealth in costs. The legacy heuristic is the equal-weight portfolio plus about one percentage point of daily turnover and 0.02 below it in Sharpe ratio. Against mean-variance the comparison reverses with the regime: mean-variance made 4 to 7× in 2019–2020 and lost money in 2021–2022, whereas RA has the smaller drawdown in 76 to 96% of the portfolios in every window, and the higher Sharpe ratio in the bear market (+0.78). In 2021–2022 both versions of the policy also beat the index, as does

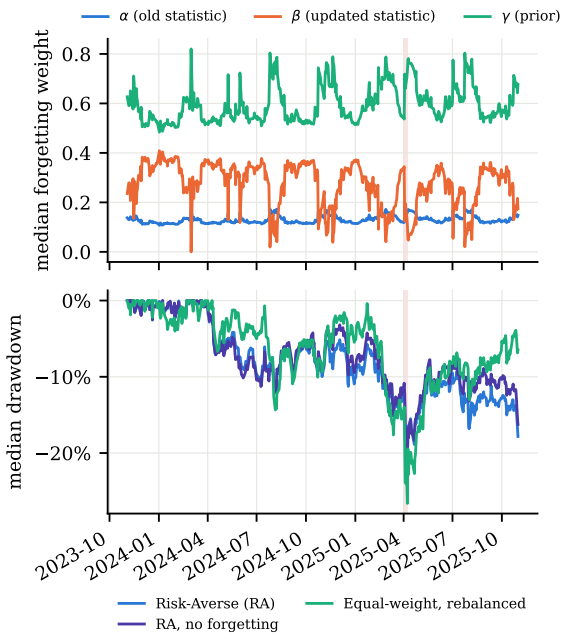


Figure S3: Top: median optimal forgetting weights over the 200 portfolios. Bottom: median drawdown of RA, of RA without forgetting and of the rebalanced equal-weight portfolio; shaded is the tariff shock of 2–9 April 2025.

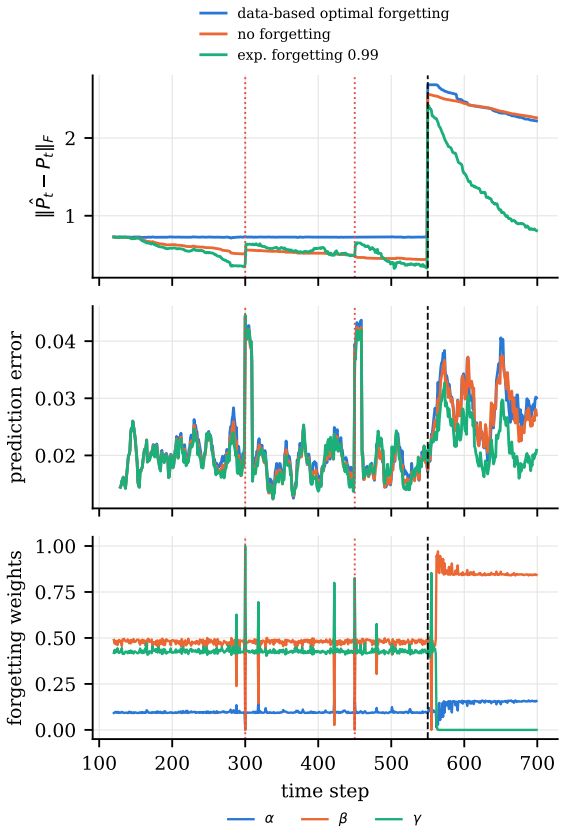


Figure S4: Synthetic stress test. Top: parameter estimation error; middle: one-step prediction error, 10-day mean; bottom: the optimal forgetting weights. Dotted: the one-day crashes; dashed: the change of all regression coefficients.

every low-turnover strategy on this universe; in the other two windows the index wins.

S10. Live Test

The live test of 1 March to 15 April 2026 (Fig. S6, Table S12) ran the pipeline on a broker feed, one portfolio, with the allocation recomputed every trading day. It returned 2.3% against 2.2% for the S&P 500 and 5.6% for the equal-weight portfolio, with the largest drawdown of the three. We keep it as a demonstration that the pipeline runs end-to-end on live data, not as evidence of performance: 32 trading days cannot separate strategies whose daily returns differ by a fraction of a percent. The 2024–2026 window of Sec. S9, which contains those dates, gives the same ranking on 100 portfolios.

S11. Corrections to the Previous Version

The experiments of this revision were run with four implementation errors corrected. They are listed here because the first of them changes all previously reported numbers.

The constrained optimiser did not optimise. The inner projected-gradient loop of the SMALBE implementation reset its iterate to the outer iterate at every iteration and stepped along the raw gradient instead of the chopped direction, so the routine returned its starting point, the softmax of the unconstrained LQR action $-\mathbf{H}_a^{-1}\mathbf{H}_s s_{t-1}$. The tables of the previous version therefore described that heuristic and not the constrained optimum. The corrected routine agrees with an independent SLSQP solver to 10^{-11} in the objective on 200 random instances, and with a brute-force search over the simplex for $N \leq 6$; the test is part of the released code. Both policies are reported throughout, the heuristic under the name “legacy softmax”.

Determinants under- and overflowed. The marginal likelihoods of the forgetting and structure criteria contain determinants of statistics of dimension above 130, which under- or overflow in double precision; they are

strategy	2019-01 – 2020-12			2021-01 – 2022-12			2023-11 – 2025-10			2024-09 – 2026-09		
	Sharpe	max DD	return	Sharpe	max DD	return	Sharpe	max DD	return	Sharpe	max DD	return
Risk-Averse (RA)	0.71	-40%	1.39	0.77	-28%	1.37	0.34	-25%	1.11	0.53	-23%	1.20
RA, legacy softmax solver	1.46*	-37%	2.27	0.96*	-23%	1.55	0.63*	-27%	1.26	1.04*	-25%	1.55
Equal-weight, rebalanced	1.48*	-36%	2.29	0.98*	-23%	1.56	0.66*	-27%	1.27	1.06*	-25%	1.56
Equal-weight, buy & hold	1.79*	-41%	3.75	0.92*	-23%	1.51	0.77*	-29%	1.36	1.01*	-26%	1.55
Inverse volatility (60d)	1.37*	-35%	1.99	1.06*	-23%	1.57	0.55*	-26%	1.20	0.91*	-25%	1.39
Min-variance (250d)	1.17*	-33%	1.70	1.01*	-24%	1.47	0.56*	-23%	1.18	0.67*	-21%	1.24
Mean-variance (250d)	1.66*	-44%	4.66	-0.01*	-37%	0.90	0.69*	-40%	1.57	0.83*	-35%	1.85
S&P 500 (SPY) buy & hold	0.93*	-34%	1.49	0.23*	-24%	1.05	1.58*	-19%	1.62	1.16*	-19%	1.42

Table S11: Mean annualised Sharpe ratio, maximum drawdown and total return over 100 random portfolios (200 for the paper window) in four 500-day test windows: 2019–2020 contains the COVID crash, 2021–2022 the 2022 bear market, 2023–2025 is the window of the previous sections, 2024–2026 lies after the paper's data. Each window uses its own 250-day warm-up. *: Sharpe ratio differs from that of the Risk-Averse policy at $p < 0.01$ (paired Wilcoxon).

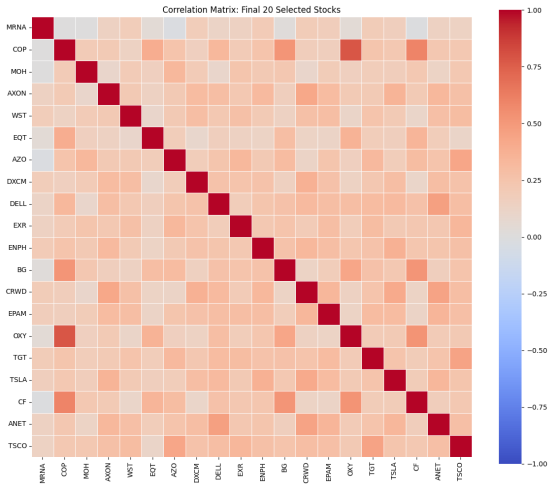


Figure S5: Correlation matrix of the 20 selected stocks of the universe of Sec. S1.

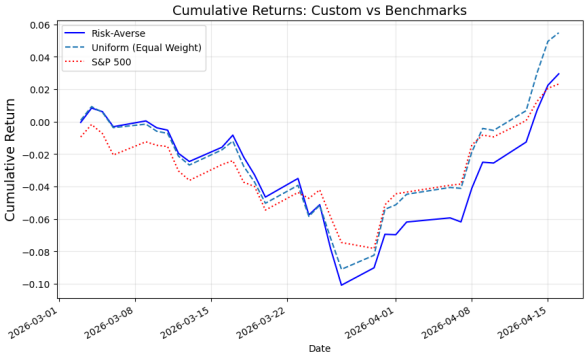


Figure S6: Returns in the live test: Risk Averse against the ideal uniform portfolio and the S&P 500 index.

replaced by log-determinants computed from the triangular square-root factors.

A wrong block in the structure criterion. The prior term of the structure log-likelihood used the posterior block $\mathbf{V}_y$ instead of the prior block $\mathbf{V}_{y,0}$.

The prior factor after structure estimation. Once the regressor set is reduced, the prior square-root factor $\mathbf{G}_0$ is recomputed from the reduced $\mathbf{V}_0$ instead of being inherited from the full one.

	RA	uniform	S&P 500
total return	1.023	1.056	1.022
max drawdown	−0.108	−0.099	−0.077
Sharpe ratio	0.074	0.133	0.069

Table S12: Live trading metrics, 1 March to 15 April 2026, one portfolio, 32 trading days.